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An adaptive (γ-Correction + Deep Learning)-based Zero-Reference Approach for Low-Light Images

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***Abstract*—Low-light images have been a severe problem, when it comes to performing surveillance and military operations at night. While, there are established computational methods, ranging from popular conventional to deep-learning methods, most methods suffers from uneven lighting conditions in an image. Therefore, a hybrid solution based on the proposed (γ-Correction + *Lite-ZeroDCE*) is introduced. It is a combination of an adaptive γ-Correction and a deep-learning-based *Lite-ZeroDCE* model to over-come the challenge of low-light image enhancement. The adaptive γ-Correction is an flexible and adaptive approach to improve the image illumination. Further, the *Lite-ZeroDCE* is based on the notion of zero-reference curve estimation. The proposed solution produces significant improvements on multiple dataset and *Lite-ZeroDCE* is light-weight (with 10.27K model parameters). Even-more, it can handle images with uneven lighting condition in an efficient manner when compared to existing methods.**

***Index Terms*—Curve estimation, Low-light image enhancement (LLIE), Zero-reference model.**

I. Introduction

The research on developing low-light image enhancement (LLIE) approaches has been the long-standing challenge in computer vision. LLIE is aimed at developing algorithm to enhance the illumination quality of images which are captured under the low-light environment. Interestingly, low-light images poses challenge during surveillance, military operations, and automobile usages at night. Low-light images suffers from poor aesthetic quality, and even lacks finer details, results into affecting the performances for image-based tasks. The causes of low-light images generation include: (i) limiting lighting conditions, (ii) poor placement of an object, and/or (iii) environmental constraints [7]. Thus, more research on the development of image enhancement techniques are needed to benefit these image-related tasks.

In regard to the low-light image enhancement (LLIE) solution, many computational approaches have been developed [1], [3], [5]–[7]. The early methods [1], [3], example using image histograms and Retinex theory [8], are primarily based on strong statistics and mathematical formulations. In contrast, deep learning based methods [5]–[7] are primarily data-dependent, with lesser need for strong statistics and mathematical knowledge. These approaches are also better in terms of performance. Further, few recent approaches even tried integrating conventional techniques (i.e., Retinex theory [8]) with deep learning [10], [11], howsoever, with few research work. Henceforth, there is still a large opportunity to work on developing a hybrid solution for LLIE.

Further, there can be more challenges that require focus, such as tackling uneven lighting conditions, where the current methods struggles to preserve natural colors and finer details. Henceforth, this paper focus on developing a low-light image enhancement (LLIE) framework. Major challenges that are dealt with include:

1) Limited research on developing a hybrid approach, i.e., to combine the conventional method with the deep-learning methods.
2) Need to develop models to handle images with uneven lighting condition.

With regard to above challenges, the key contributions are as follows:

1) The proposed solution utilizes an adaptive γ-correction and CNN-based "*Lite-ZeroDCE*" architecture to generate well-lit images.
2) Moreover, "*Lite-ZeroDCE*" is a lighter model (with $\approx$ 10.27K parameters) and is based on estimating the light enhancement curves.
3) Experiments with multiple datasets demonstrate the superiority of the proposed (γ-correction + *Lite-ZeroDCE*) method. It outperforms LIME [3], ZeroDCE [5] and ZerDCE++ [6]. In addition, it manged images with the uneven lighting conditions in a better way.

The rest of sections are: Section II is the related works. Section III and Section IV discuss the datasets and the pro-

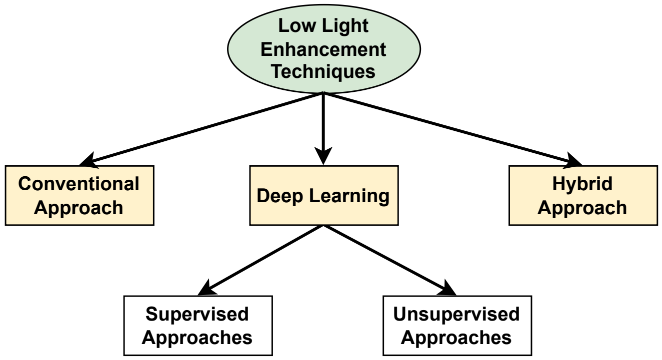

Fig. 1: A generalized hierarchical classification of existing methods.

posed solution, respectively. Section V is on the experimental results. Section VI lastly concludes the paper.

II. Related Work

The current literature on low-light enhancements, which is shown in Fig. 1, can be grouped into: (i) *Conventional approaches* [1], [3], (ii) *Deep learning based approaches* [2], [5]–[7], and (iii) *Hybrid approaches* [9]–[11]. Early or conventional approaches for low-light enhancements [1], [3], are based on statistical and mathematical formulations. Li *et al.* [1] introduced a solution based on the illumination map estimation. The Retinex theory [8] is employed over the images to decomposed them into sub-components, i.e., illumination and reflectance. Further, LIME [3] is based on the (illumination map + γ-correction) technique.

Next major category include approaches based on deep-learning which can be further sub-grouped into: (i) *supervised-based*: require paired (low, high)-light image dataset [4], and (ii) *unsupervised-based*: require only low-light image [5]–[7]. Guo *et al.* [5] introduced *ZeroDCE*, that used CNN-based U-Net structure for the unsupervised curve estimation. Next, Li *et al.* [6] introduced *ZeroDCE++* that is a lighter generalization of ZeroDCE [5] (having $\approx 10K$ model parameters). In contrast, ZeroDCE [5] has $\approx 80K$ model parameters. Further, Mu *et al.* [7] proposed ZeroDCE-Tiny which is more lighter model with $\approx 3K$ model parameters.

Lastly, there are hybrid approaches [9]–[11] that merged the conventional Retinex theory [8] and the deep-learning technique. Hu *et al.* [9] used pre-enhanced low-light images (using Retinex method) to train the refinement neural network. Then, Tang *et al.* [10] introduced (enhancement network + de-noising network) to generate well-lit images. The enhancement network is based on CNNs and Retinex theory [8]. In an another approach, Jiang *et al.* [11] proposed a Retinex theory [8] based end-to-end U-Net architecture.

III. Datasets

The validation of the proposed approach is performed using two different low-light image datasets. The first dataset is the UnLOL dataset (https://drive.google.com/drive/folders/1fbzfpK0CfPA8nGMu-wbx0nXnL6g_Gh2Q?usp=sharing), which is diverse and include images with more uneven lighting conditions [12]. This has a total of 128 images for training and 43 images for testing. The second dataset is LOL dataset [2]. However, it lacks enough diversity in the lighting condition. The LOL dataset has 485 samples for training and 15 sample for testing.

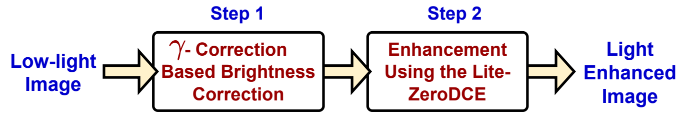

Fig. 2: Flow diagram demonstrating major steps with the proposed solution.

IV. Proposed Method

The proposed work is a hybrid method for enhancing the low-light images using the γ-correction and the deep-learning based "*Lite-ZeroDCE*" architecture. *Lite-ZeroDCE* (elaborated in Section IV-B) is based on the notion of zero-reference that relaxes the need for the paired images during the training. The different steps with the proposed work are shown in Fig. 2, which are elaborated next:

A. Applying Adaptive Gamma Correction Algorithm

The low-light image (I) undergoes pre-processing step using the adaptive γ-correction algorithm as discussed in Algorithm 1. The algorithm has three major steps as outlined in Algorithm 1 and are discussed next.

Algorithm 1 Adaptive Gamma Correction

Inputs: Low-light image (I)
Output: Well-lit image (I')

1: Compute the average intensity of an image.

$$Avg.\ Int = np.mean(I) \tag{1}$$

2: Set the gamma parameter (γ) based on the average intensity of an image.

$$\gamma = \begin{cases} 0.6, & \textbf{if } Avg.\ Int < 70 \\ 0.8, & \textbf{if } 70 \leq Avg.\ Int < 100 \\ 1.0, & \textbf{if } 100 \leq Avg.\ Int \leq 150 \\ 1.2, & \textbf{if } 150 < Avg.\ Int \leq 180 \\ 1.4, & \textbf{if } 180 < Avg.\ Int \end{cases} \tag{2}$$

3: $I' = I^{\gamma}$ # Apply gamma correction
4: **return** I'

The first step is computing the avg. intensity of the low-light image, denoted with $Avg.\ Int$. It is a measure of overall brightness of an image. Then, in the second step, the gamma parameter (γ) is chosen based the average intensity of the image. The γ value is kept small for a low-light image with very poor overall brightness. In contrast, the γ value is kept large, if the overall brightness of an image is high. This is represented using Equation 2. Lastly, in the third step, the

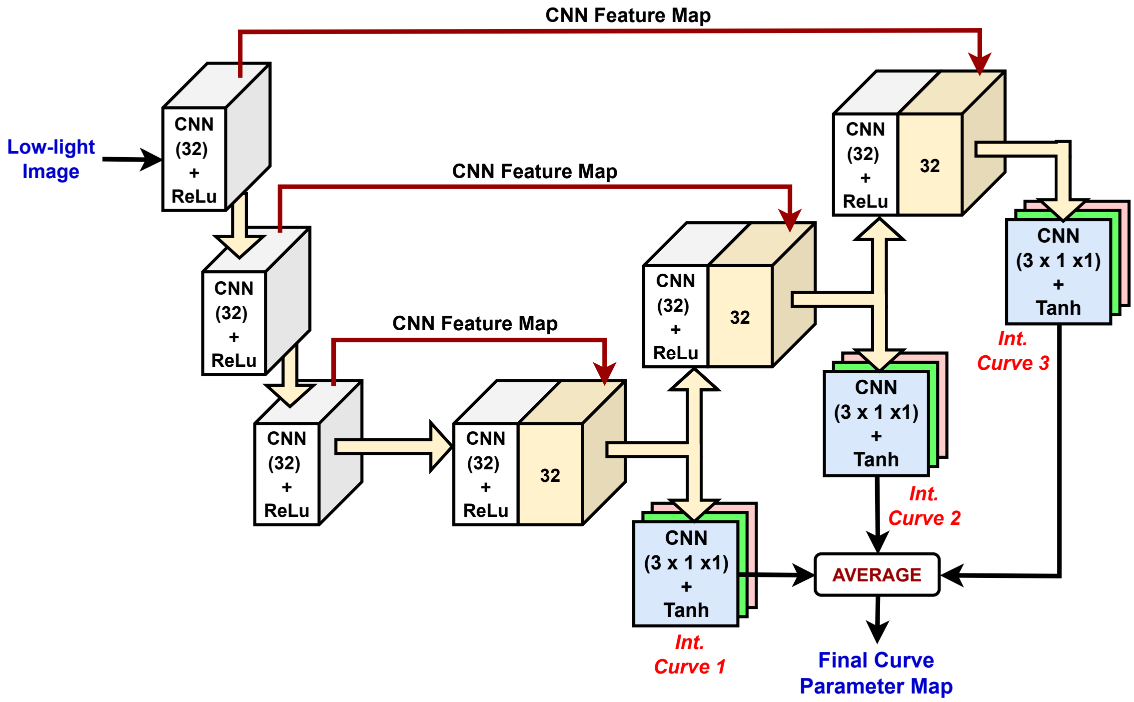

Fig. 3: Proposed architecture of the *Lite-ZeroDCE* model. This utilizes depth-wise separable CNNs to realize the light-weight architecture and is based on the notion of estimation of multiple curve parameters.

chosen γ-correction is applied which is a non-linear operation on the low-light image. It helps enhance the brightness of an image, where the gamma corrected image is denoted as I'.

The advantage of the adaptive γ-correction algorithm is that it helps dynamically adjust the brightness of an image based on the average brightness quality of the low-light image.

B. Training a Lite-ZeroDCE Model

The images produced from the γ-correction step (discussed in Section IV-A) are next utilized for training the proposed *Lite-ZeroDCE* model. As shown in Fig. 3, it is a light-weight neural network architecture that uses a depth-wise separable CNN. The working of *Lite-ZeroDCE* model is based on "*Zero-Reference Deep Curve Estimation*" that allows to improve the illumination conditions of a low-light image. Note, "*Zero-Reference*" symbolizes relaxation from the requirement of pair of low and high light images that helps generalize well for the new unseen data.

The *Lite-ZeroDCE* model can be mathematically formulated as: Let, the gamma-corrected input image be $I'(x)$ and the enhanced image be $I''(x)$, where x denote an image pixel. Then, $I''(x)$ is produced using Eq. 3:

$$I''(x) = I'(x) + wI'(x) * [1 - I'(x)] \tag{3}$$

where, $w \in [-1, +1]$ is a learnable curve parameter.

For a dynamic adjustment, the Eq. 3 is applied in an iteration (denoted with n) which is formulated as:

$$I''_n(x) = I'_{n-1}(x) + w_n I'_{n-1}(x)[1 - I'_{n-1}(x)] \tag{4}$$

1) Lite-ZeroDCE Model Architecture: *Lite-ZeroDCE*, as shown in Fig. 3, is a U-net based architecture that utilizes nine depth-wise separable CNNs for the curve-estimation. The depth-wise separable CNNs is based on two major operations, which are: i) *depth-wise convolution*: where each channel is processed separately, and ii) *point-wise convolution*: each pixel is processed along the channels.

The first six CNN layers are used for processing the image. These CNN layers have following hyper-parameters: no. of filters = 32, filter-size (3 x 3), stride = 1, and activation = *ReLU*. The remaining three CNN layers, as shown in Fig. 3, are used to learn the intermediate curve parameters, denoted with, *Int. Curve* 1, *Int. Curve* 2, and *Int. Curve* 3. Individual intermediate curve parameters helps understand different levels of abstraction for low-light images. These CNN layers have following hyper-parameters: no. of filters = 3, filter-size (1 x 1), stride = 1, and activation = *Tanh*.

2) Non-Referential Loss Function: The loss function with the ZeroDCE++ [6] model is used for this work. Mathematically, the loss function is:

$$\begin{aligned} Loss = w_1 * L(col.\ constancy) + w_2 * L(exp.) + \\ w_3 * L(illu.\ smoothness) + w_4 * L(spa.\ constancy) \end{aligned} \tag{5}$$

where, $w_1 = 2$, $w_2 = 10$, $w_3 = 100$ and $w_4 = 5$.

1) **Color constancy loss** [6]: Ensure to retain the true

colors.

$$L(col.\ constancy) = \sum_{\forall(C_1,C_2)} (I_{avg}(P) - I_{avg}(Q))^2 \tag{6}$$

where, I_{avg} is avg. intensity and $C_1, C_2 \in \{R, G, B\}$.

2) **Exposure loss** [6]: Improves the under-exposed/over-exposed regions.

$$L(exp.) = \frac{1}{M} \sum_{k=\{1,2,...,M\}} |I^k_{avg} - E| \tag{7}$$

where, I^k_{avg} is the avg. intensity of the local region, and $E = 0.6$ is the well-exposedness level.

3) **Illumination smoothness loss** [6]: Ensures retain the monotonicity relationships bet. the neighboring pixels.

$$L(illu.\ smoothness) = \frac{1}{N} \sum_{n=1}^{N} \sum_{\forall C} (|\nabla_x A^C_n| + |\nabla_y A^C_n|)^2 \tag{8}$$

where, N denote no. of iterations, $C \in \{R, G, B\}$, ∇_x denote gradient along x-axis, ∇_y denote gradient along y-axis and A denote the curve parameter.

4) **Spatial consistency loss** [6]: Helps preserve the local spatial relationships of the enhanced image, where R denote count of image sub-regions.

$$L(spa.\ constancy) = \frac{1}{R} \sum_{r=1}^{R} \sum_{j \in \Omega(r)} (|I'_r - I'_j| - |I_r - I_j|)^2 \tag{9}$$

where, I and I' represents the avg. intensity of sub-region in the input image and its enhanced version, respectively.

3) Model hyper-parameters:

- Input image size = (256 x 256),
- Optimizer = *adam* (lr = 0.0001),
- No. of epochs = 200, and
- Batch-size = 16.

V. Result and Discussion

In this section, the proposed (auto-contrast + *I-ZeroDCE++*)-based framework is evaluated using the dataset discussed in Section III.

A. Evaluation Metric

Given, low-light/enhanced images (say I_1) and well-lit images (say I_2), metrics such avg. PSNR, avg. SSIM, and avg. MS-SSIM [6] are used to evaluate the proposed work. These are computed as:

$$PSNR(I_1, I_2) = 10log_{10}\left(\frac{R^2}{Mean\ Sq.\ Err.(I_1, I_2)}\right) \tag{10}$$

where, *R* is the maximum value of pixels in the high-quality image (I_1).

TABLE I: Ablation analysis: Comparisons on the Proposed Dataset

S. No.	Color constancy loss weight	Average PSNR (↑)	Average SSIM (↑)	Avg. MS-SSID (↑)
1.	w_1 = 1	**13.5887**	**0.6069**	**0.7138**
2.	w_1 = 2	13.5791	0.6065	0.7137
3.	w_1 = 5	13.5703	0.6063	0.7136
4.	w_1 = 10	13.5682	0.6062	0.7136
5.	w_1 = 20	13.5655	0.6060	0.7135

TABLE II: Ablation analysis: Comparisons on the LOL Dataset (Model = γ-Correction + *Lite-ZeroDCE*)

S. No.	Color constancy loss weight	Average PSNR (↑)	Average SSIM (↑)	Avg. MS-SSID (↑)
1.	w_1 = 1	18.0997	0.6207	0.8766
2.	w_1 = 2	**18.1321**	0.6212	**0.8766**
3.	w_1 = 5	18.1121	0.6218	0.8765
4.	w_1 = 10	18.0197	0.6224	0.8759
5.	w_1 = 20	17.8908	**0.6232**	0.8754

$$SSIM(I_1, I_2) = \frac{(2\mu_{I_1}\mu_{I_2} + c_1)(2V_{I_1I_2} + c_2)}{(\mu^2_{I_1} + \mu^2_{I_2} + c_1)(V^2_{I_1} + V^2_{I_2} + c_2)} \tag{11}$$

where, $(\mu_{I_1}, V^2_{I_1})$ and $(\mu_{I_2}, V^2_{I_2})$ represent (mean, variance) for I_1 and I_2, respectively.

$$MS{-}SSIM(I_1, I_2) = [L_M(I_1, I_2)]^{a_M} \prod_{j=1}^{M} [SSIM_j(I_1, I_2)]^{b_j} \tag{12}$$

where, M is the number of scales, $L_M(I_1, I_2)$ is the luminance similarity, and (a_M, b_j) are parameters signifying weights for different scales.

B. Evaluating The Impact Of Color Constancy Loss

In this sub-section, the impact of color constancy loss with the proposed (γ-Correction + *Lite-ZeroDCE*)-based solution is evaluated. The weight corresponding to the color constancy loss is varied between {1, 2, 5, 10, and 20} for deeper analysis. The results for the same are given in Table I (for the proposed dataset) and Table II (for the LOL dataset).

Clearly, results suggest a better performance with the lower weight parameter for the color constancy loss. For the proposed dataset, the optimal results are obtained with the weight value (w_1) equals to 2, while for the LOL dataset, the optimal results are obtained with the w_1 equal to 1. Furthermore, increasing the weight parameter, however, causes decline in the results.

Overall, a weight value (w_1) equals to 2 gives good results, and hence are used for further analysis.

C. State-Of-The-Art (SOTA) Comparisons

A comparison between the proposed (γ-Correction + *Lite-ZeroDCE*) solution and the current SOTA methods is presented

TABLE III: SOTA Comparisons with respect to both the Datasets

S. No.	Dataset $\longrightarrow$ Method	Proposed Dataset Average PSNR (↑)	Proposed Dataset Average SSIM (↑)	Proposed Dataset Avg. MS-SSID (↑)	LOL Dataset Average PSNR (↑)	LOL Dataset Average SSIM (↑)	LOL Dataset Avg. MS-SSID (↑)
1.	Low Quality Images	10.3573	0.4581	0.6584	7.7733	0.1952	0.5017
2.	Adaptive γ-Correction Method	12.4037	0.5730	0.7060	10.6168	0.4764	0.7295
3.	LIME ($\alpha = 0.2, iteration = 10$) [3]	12.7504	0.5564	0.6737	15.8437	0.4895	0.8415
4.	ZeroDCE [5]	11.9265	0.5552	0.7029	16.2001	0.5692	0.8747
5.	ZeroDCE++ [6]	11.9143	0.5546	0.7040	16.3754	0.5698	0.8755
6.	Auto-contrast + I-ZeroDCE++ [12]	12.4104	0.5683	0.7067	15.8833	0.5164	0.8473
7.	γ-Correction + *Lite-ZeroDCE*	**13.5791**	**0.6065**	**0.7137**	**18.1321**	**0.6212**	**0.8765**

TABLE IV: Parameter comparisons for different methods

S. No.	Method	Method Parameter
1.	Low Quality Images	–
2.	Adaptive γ-Correction Method	–
3.	LIME ($\alpha = 0.2, iteration = 10$) [3]	–
4.	ZeroDCE [5]	79.42K
5.	ZeroDCE++ [6]	10.27K
6.	Auto-contrast + I-ZeroDCE++ [12]	11.81K
6.	γ-Correction + *Lite-ZeroDCE*	**10.27K**

in this section. SOTA methods that are used for the comparison include LIME [3], ZeroDCE [5], ZeroDCE++ [6], and Auto-contrast + I-ZeroDCE++ [12]. Besides, results with the adaptive γ-correction approach are also reported.

1) Objective Evaluation: This is based on popular metrics discussed in Section V-A, and are shown in Table III. In addition, Table IV presents the comparative analysis based on the number of model parameters.. As observed from Table III, it is very clear that the proposed (γ-Correction + *Lite-ZeroDCE*)-based solution is superior to other. It is worth noting that the adaptive γ-Correction methods helped to improve the illumination quality, though the (γ-Correction + *Lite-ZeroDCE*) model performed much better as observed from Table III. Using the proposed dataset, the margin of improvements when compared to the LIME [3] are: +0.8287 (6.50%), +0.0501 (9.00%), and +0.040 (5.94%) for avg. PSNR, avg. SSIM, and avg. MS-SSIM, respectively. The corresponding numbers with respect to LOL dataset are: +2.2584 (14.25%), +0.1317 (26.90%), and +0.035 (4.16%), respectively. A very similar kind of improvements are observed when compared to ZeroDCE [5], ZeroDCE++ [6] and Auto-contrast + I-ZeroDCE++ [12].

Furthermore, a look at the number of parameters (shown in Table IV), the proposed (γ-Correction + *Lite-ZeroDCE*)-based solution exhibits a lowest number of parameter, which is also equivalent to ZeroDCE++ [6], i.e., 10.27K parameters.

2) Subjective Evaluation Based on Sample Test Images: The image enhancements using the proposed and other SOTA methods for few sample test images are discussed here from the human perspective. The enhanced images are illustrated in Fig. 4 and Fig. 5 for the proposed dataset [12] and LOL dataset [2], respectively.

For the proposed dataset [12], the illustrations in Fig. 4

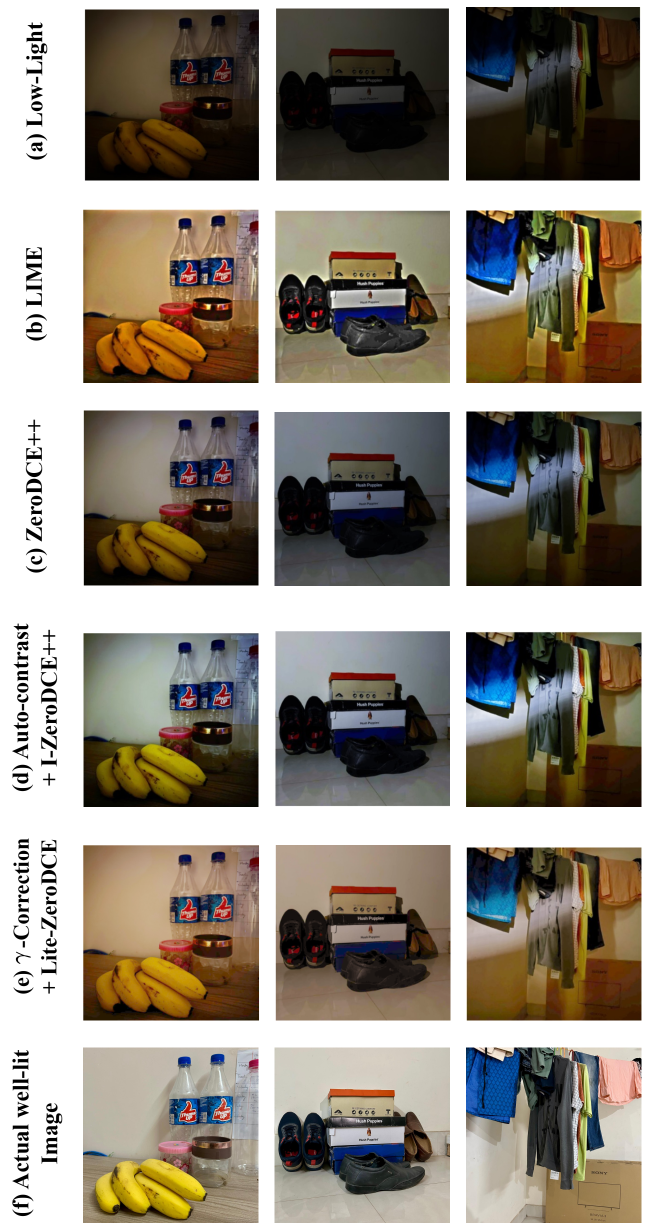

Fig. 4: Image enhancements for sample test images obtained for the proposed dataset.

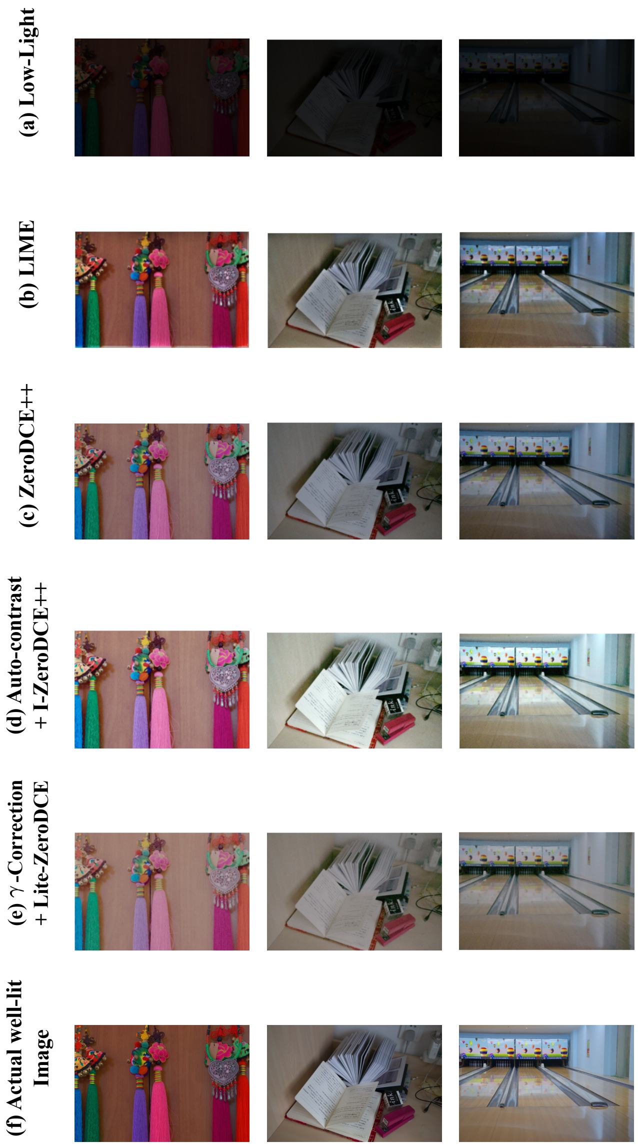

Fig. 5: Image enhancements for sample test images obtained for the LOL dataset.

shows that well-lit images are produced with the LIME [3] method, though, the enhanced images tends to contain a lot of noise, i.e., poorly defined coarse edges. In contrast, test images enhanced with the proposed (γ-Correction + *Lite-ZeroDCE*) solution are better when compared to the other SOTA methods. This includes comparisons with the LIME [3] method as well as other known methods such as ZeroDCE [5] and ZeroDCE++ [6]. Similar results are obtained for images from the LOL dataset as well (as observed from Fig. 5).

VI. Conclusion

Low-light images have been challenging, when it comes to performing surveillance and military operations at night. While, there are established methods, ranging from conventional to deep-learning methods, most methods suffers from uneven lighting conditions in an image. Therefore, a hybrid solution based on the combination of an adaptive γ-Correction + *Lite-ZeroDCE* is introduced. The adaptive γ-Correction method support adaptive image enhancements based on the image mean intensity. Then, deep-learning based *Lite-ZeroDCE* is used to further improve the lighting conditions of an image. The proposed solution produces significant improvements on multiple dataset and is even light-weight (with 10.27K parameters). Even-more, it can handle images with uneven lighting condition in a better way when compared to existing methods. Future work will investigate the proposed models for larger dataset and explore it for other domain.

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